

Theory of Mechanism and Machine

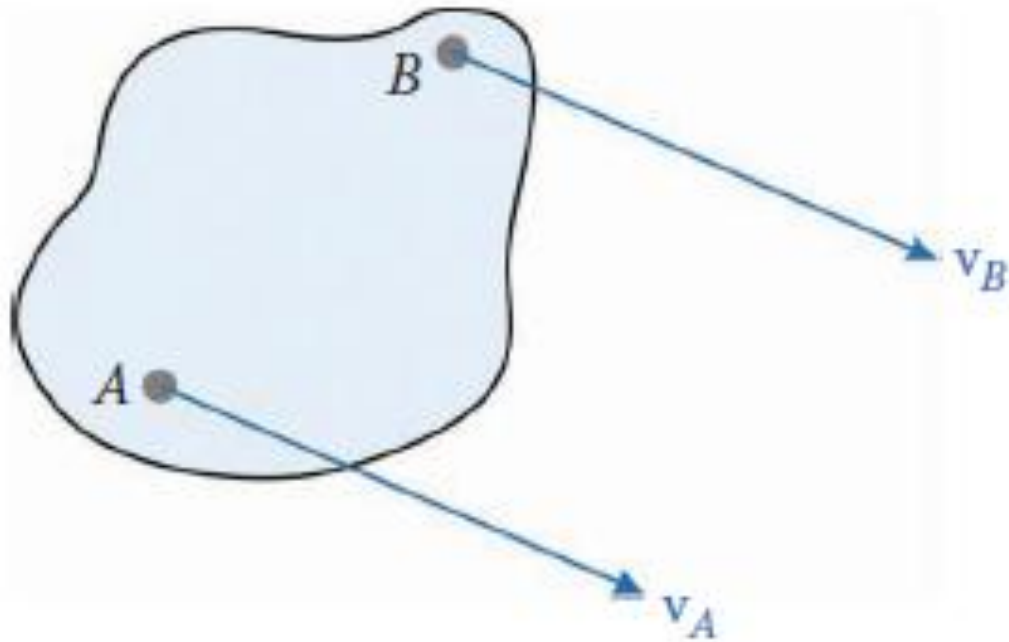
(Week-4)

Velocity Analysis

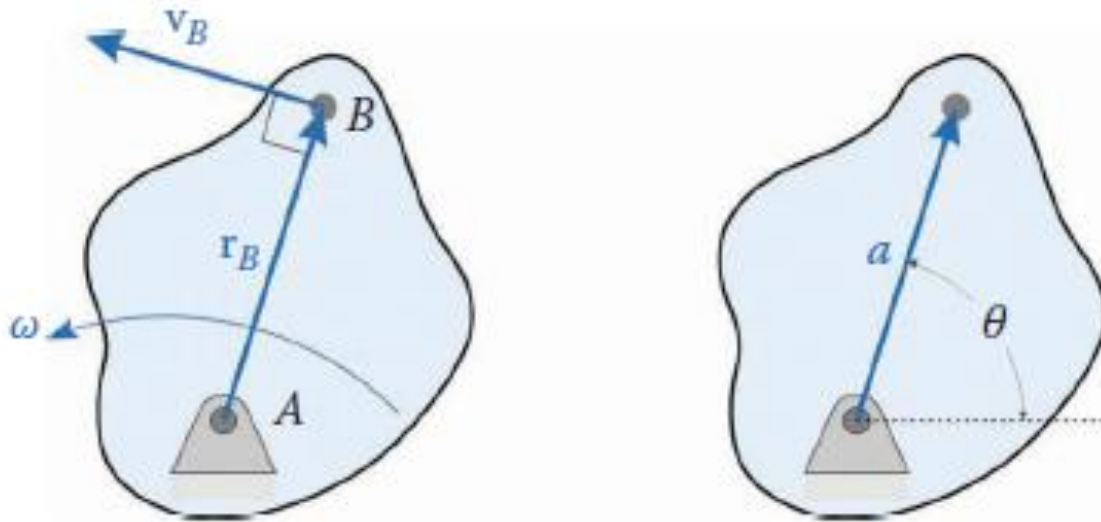
- *The velocity analysis is necessary for determining the acceleration of the points in the mechanism. The choice of the method depends upon the nature of the mechanism and accuracy required.*
- There are two methods to determine the velocities of links of mechanisms.
 1. Instantaneous centre method.
 2. Relative velocity method.

- ***Instantaneous Centre Method:***
- *This method is convenient and easy to apply in simple mechanism. Before discussing this method, let us consider the following types of motion that a body can have :*
 - (i) Motion of translation.*
 - (ii) Motion of rotation.*
 - (iii) Combination of motion of translation and motion of rotation.*

- In the case of pure translation, all points on the body move with equal speed, and in the same direction. Thus, as shown in Figure the points A and B have equal velocity. $v_B = v_A$

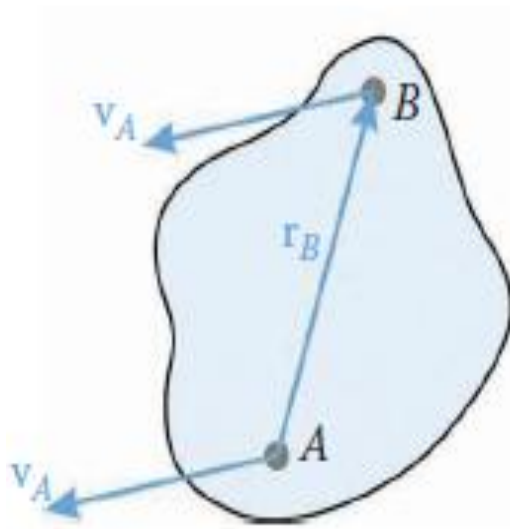


- The figure shows a rigid body pinned to the ground at point A. Because the body is pinned to the ground, it can only experience pure rotation, and the point B traces a circular arc centered at point A. The velocity vector, v_B , is tangential to the circle, which means that it is perpendicular to the vector r_B .

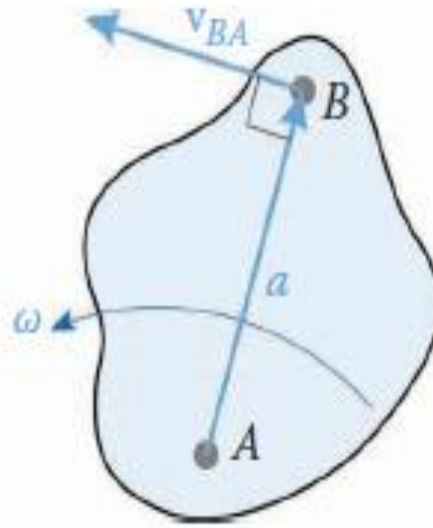


$$\mathbf{r}_B = a\mathbf{e} = a \begin{Bmatrix} \cos \theta \\ \sin \theta \end{Bmatrix}$$

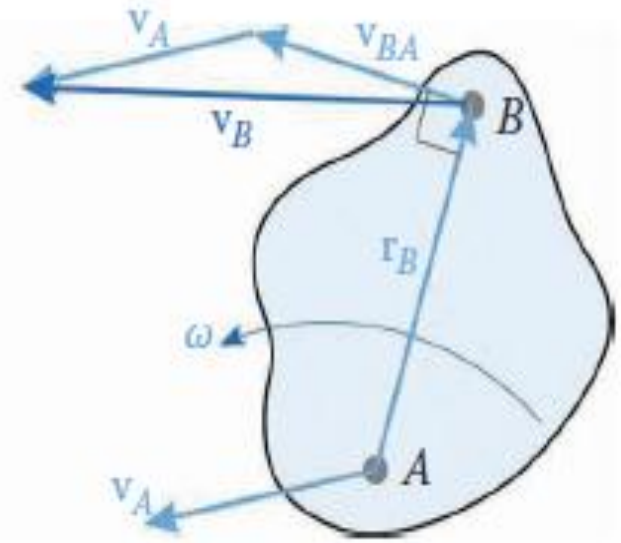
- Any motion can be described as a combination of rotational and translational movements. Consider the motion of the rigid body in Figure . The point A moves with velocity v_A . Since the body is rigid, the point B must also share this velocity. The rotation of the body also causes the point B to move relative to A in the same manner as for pure rotation.



$$|\mathbf{v}_{BA}| = a\omega$$



$$\mathbf{v}_{BA} \perp \mathbf{r}_B$$



$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{BA}$$

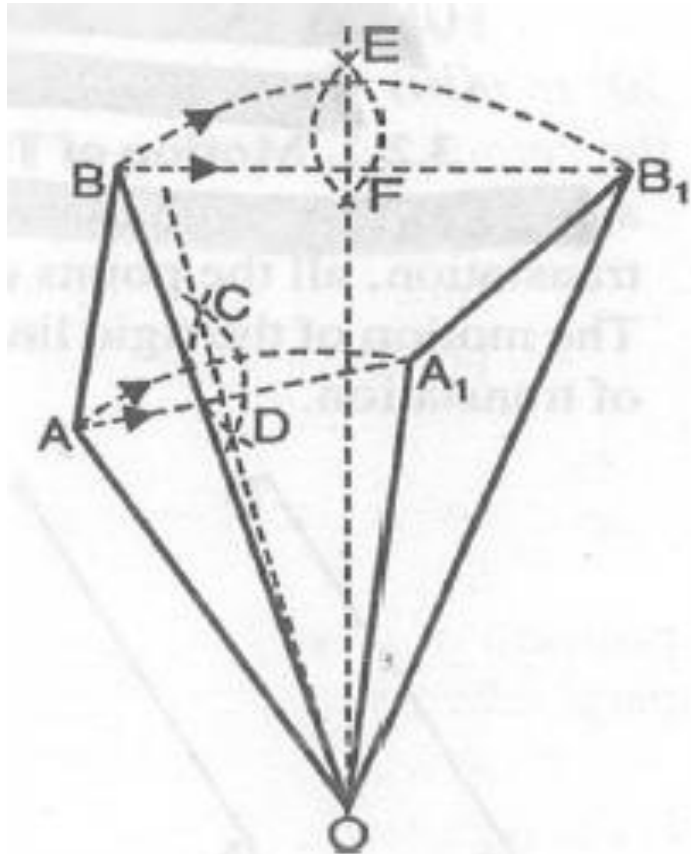
A link as a whole may be considered to be rotating about an imaginary centre or about a given centre at a given instant. Such a centre has zero velocity, i.e. the link is at rest at this point. This is known as the instantaneous centre or centre of rotation. This centre varies from instant to instant for different positions of the link. The locus of these centres is termed the centrode.

Any three bodies in a plane will have exactly three instant centers and all three will lie in a straight line.

- Since there exists one instant center (IC) per pair of bodies, we may calculate the total number of ICs(C) with the following formula.*

$$C = \frac{n(n-1)}{2}$$

- *Location of the Instantaneous Centre:*
- *The position of the instantaneous centre of rotation is determined as given below:*
- *Let the link AB in a short interval of time changes its position from AB to A₁B₁. The point A of the link has moved to point-A₁ as shown in Fig , where the point B of the link has moved to the point -B₁ as:*



Draw the right bisectors of chord AA₁ and chord BB₁ .

Let CD is the right bisectors of AA₁ whereas EF is the right bisector of BB₁ Let these two bisectors meet at the point O. Then the point O is the instantaneous centre of rotation of the link AB.

This means the link AB as a whole has rotated about O.

- Let V_A = Linear velocity of point A
- V_B = Linear velocity of point B
- ω = Angular velocity of link AB about O.
- Then linear velocity = Angular velocity $\times$ r.
- Linear velocity of point A is given by
- V_A = Angular velocity of point A about O $\times$ Distance of A from centre of rotation (i.e., from O) = $\omega \times AO$

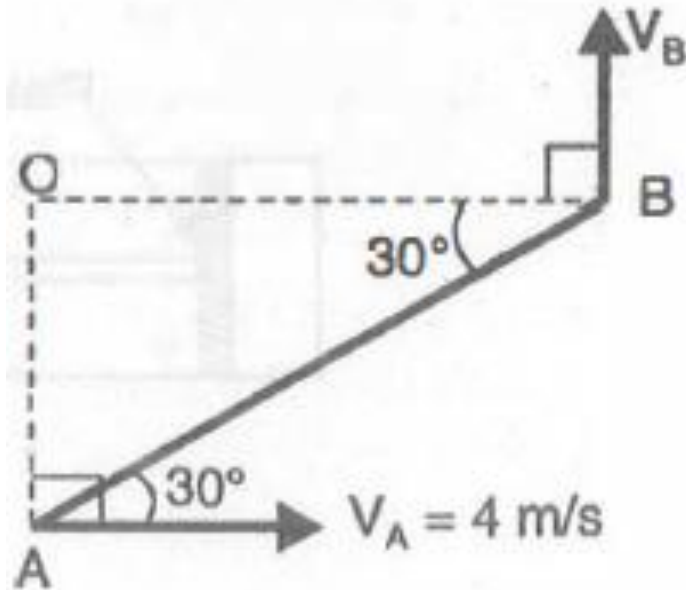
$$\omega = \frac{V_A}{AO}$$

$$\text{Similarly : } \omega = \frac{V_B}{BO} = \frac{V_C}{CO} = \omega$$

$$\text{Then; } \frac{V_A}{AO} = \frac{V_B}{BO} = \frac{V_C}{CO} = \omega$$

Q 1: A link AB is moving in a vertical plane. At a certain instant, when the link is inclined at 30° to the horizontal, the point A is moving horizontally at 4 m/s, while B is moving vertically upwards. Find the velocity of B.

Draw the FBD:



$$\frac{V_A}{V_B} = \frac{AO}{BO}$$

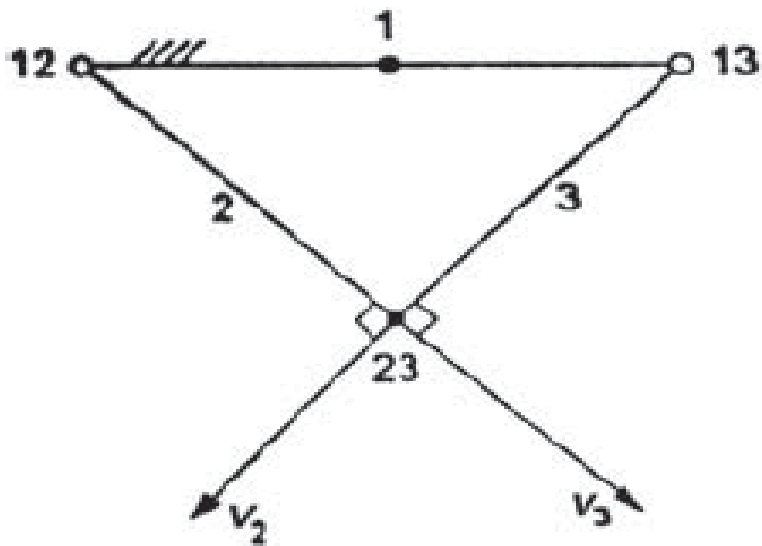
$$\text{In } \triangle AOB, \tan 30^\circ = \frac{AO}{BO}$$

$$\frac{4}{V_B} = \frac{1}{\sqrt{3}}$$

$$V_B = 4 \times \sqrt{3} = 4 \times 1.732 = 6.928 \text{ m/s.}$$

Arnold-Kennedy Theorem:

This theorem states that if three-plane bodies have relative motion among themselves their three instantaneous centers must lie on a straight line.



Proving three-centres theorem

Consider three rigid links 1, 2, and 3; 1 being a fixed link. 12 and 13 are the instantaneous centres of links 1, 2 and 1, 3 respectively.

Let 23 be the instantaneous centre of links 2, 3, lying outside the line joining 12 and 13.

The links 2 and 3 are moving relative to link 1.

Therefore the motion of their instantaneous centre 23 is to be the same whether it is considered in body 2 or 3.

If the point 23 is considered on link 2, then its velocity v_2 is perpendicular to the line joining 12 and 23.

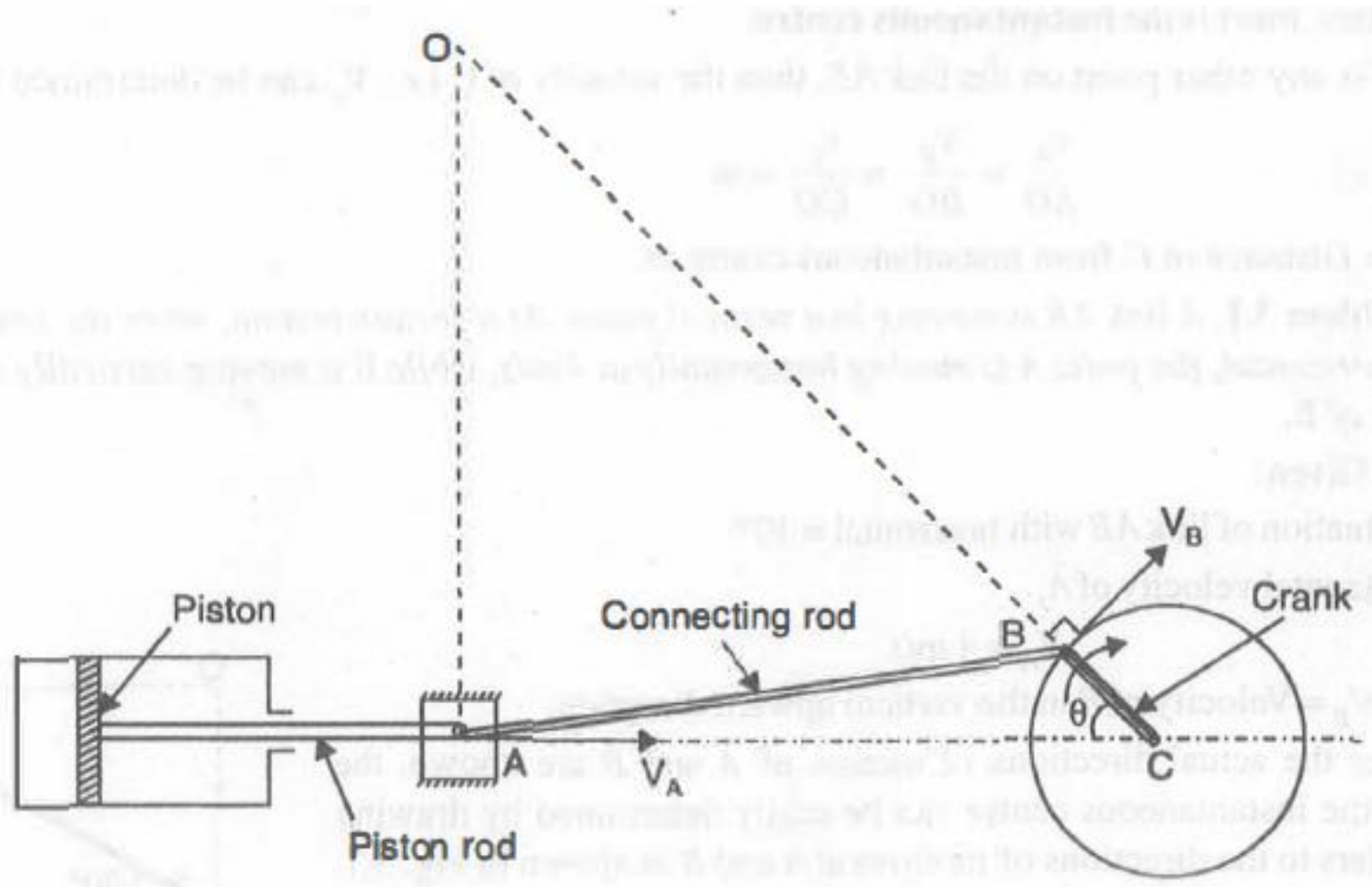
If the point 23 lies on link 3, then its velocity v_3 must be perpendicular to the line joining 13 and 23.

The velocities v_2 and v_3 of point 23 are in different directions, which is not possible.

The velocities v_2 and v_3 of instantaneous centre 23 will be equal only if it lies on the line joining 12 and 13.

Hence all the three instantaneous centres 12, 13 and 23 must lie on a straight line.

Analysis of reciprocating engine mechanism by instantaneous centre method:



The figure shows a reciprocating engine mechanism in which AB is the connecting rod and BC is the crank. The crank BC is rotating at a uniform angular velocity in the clockwise direction about point C. The point A connected to the piston rod and connecting rod is having to and fro motion in the horizontal plane. The connecting rod is having a combined motion of translation and rotation

N =Revolution of crank in r.p.m, ω = Angular velocity of crank
 r =Radius of crank, i.e., length BC, L =Length of connecting rod AB.

$$\omega = \frac{2\pi N}{60}$$

$$\text{Linear velocity of point B} = V_B = \omega \times BC = \omega \times r \quad (i)$$

The position of the instantaneous centre can be determined by drawing perpendiculars to the directions of the velocities at A and B. The two perpendiculars meet at O, which is the instantaneous centre for the connecting rod AB.

ω_{AB} = Angular velocity of the connecting rod AB about O.

$$V_A = \omega_{AB} \times AO \quad (ii)$$

$$V_B = \omega_{AB} \times BO \quad (iii)$$

Equating equations (i) and (iii), we get

$$\omega \times r = \omega_{AB} \times BO$$
$$\omega_{AB} = \frac{\omega \times r}{BO}$$

Similarly

$$V_A = \frac{\omega \times r}{BO} \times AO$$

Q 2: The crank of a reciprocating engine is rotating at 210 r.p.m. The lengths of the crank and connecting rod are 20 cm and 100 cm respectively. Find the velocity of the point A (i.e., velocity of piston), when the crank has turned through an angle of 45° with the horizontal. Also locate the IC.

Rotation of crank,

$$N = 210 \text{ r.p.m.}$$

$\therefore$ Angular velocity of crank,

$$\omega = \frac{2\pi N}{60} = 2 \times \frac{22}{7} \times \frac{210}{60} = 22 \text{ rad/s}$$

Crank length or radius,

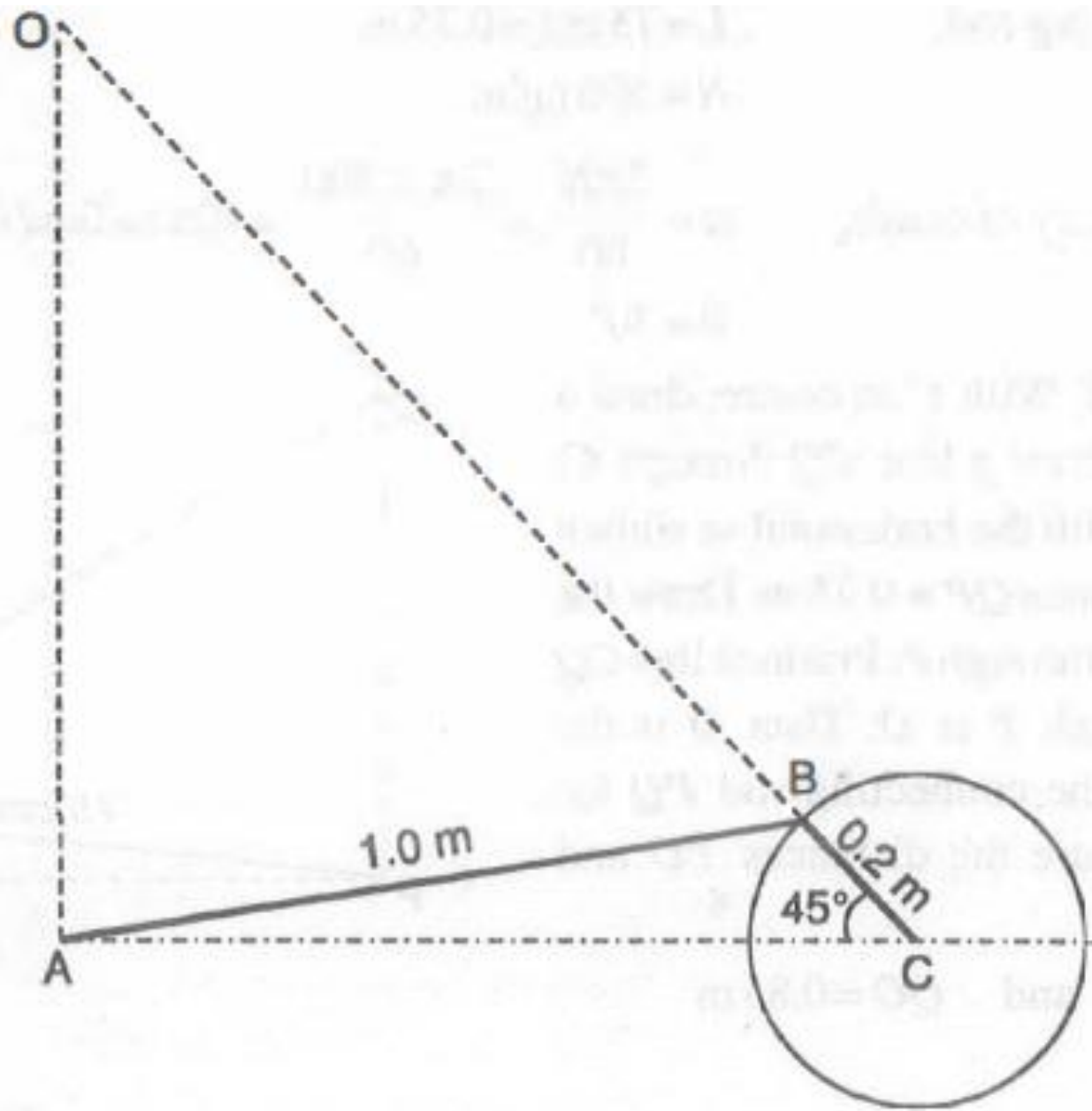
$$r = 20 \text{ cm} = 0.20 \text{ m}$$

Length of connecting rod,

$$L = 100 \text{ cm} = 1.0 \text{ m}$$

Angle turned by crank,

$$\theta = 45^\circ.$$



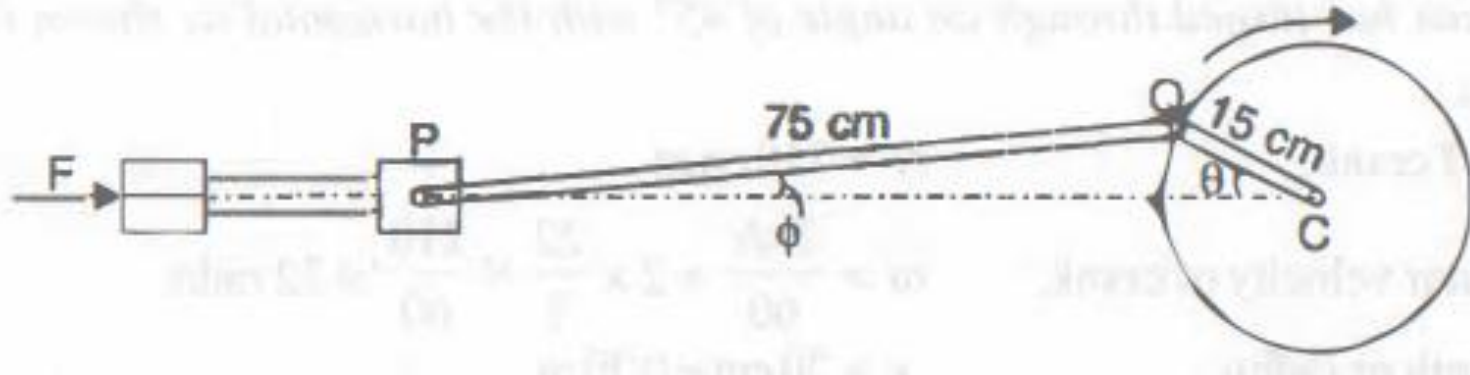
- *The procedure of finding velocity graphically is:*
- *1 . With C as centre, draw a circle of radius 0.2 m.*
- *2. Draw a line BC through C, making an angle of 45° with the horizontal and cutting the circle at B.*
- *3. Draw a horizontal line AC through C.*
- *4. With B as centre, cut off AB equal 1 .0 m.*
- *5. Produce the line CB. And draw a normal to the line AC passing through A. The intersection of the normal through A and produced line BC, gives the instantaneous centre O for the rod AB.*

Measuring lengths AO and BO from the diagram, we get AO=1.15m and BO=1.41m.

$$V_A = \frac{\omega \times r \times AO}{BO} = \frac{22 \times 0.2 \times 1.15}{1.41} = 3.58 \text{ m/s.}$$

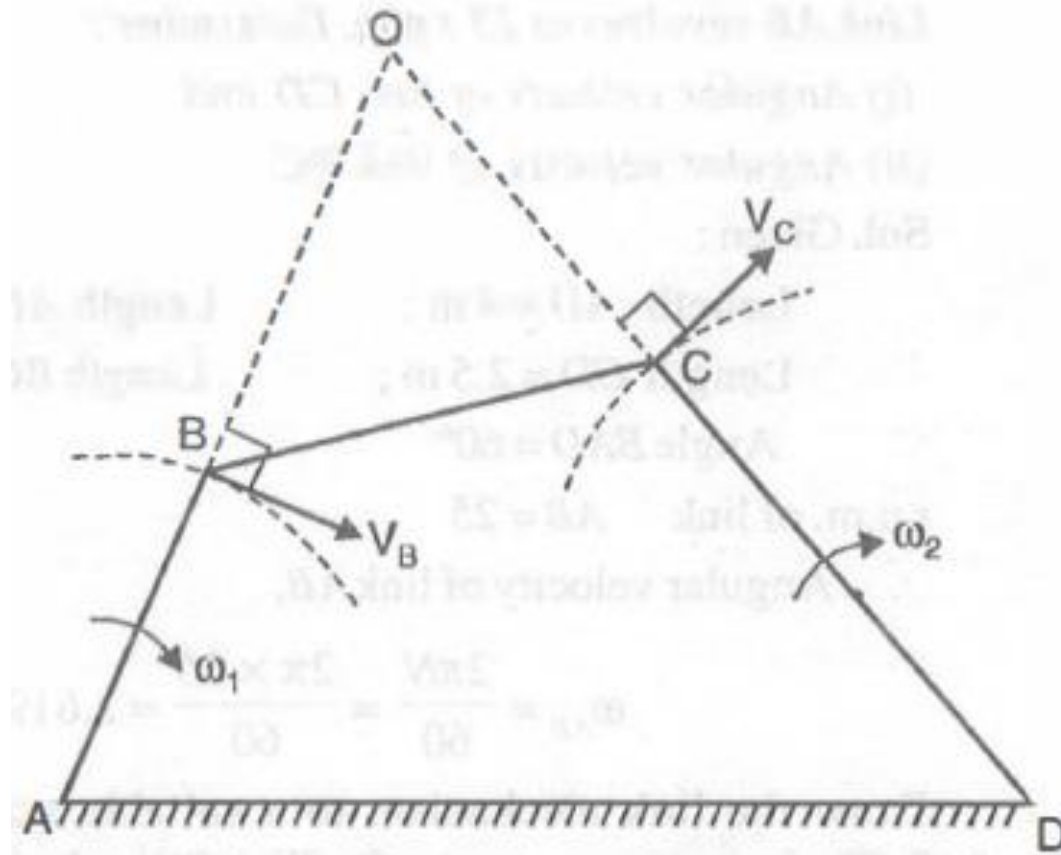
Home Work:

Q 2: A steam engine has a crank radius of length 15 cm and connecting rod of length 75 cm as shown in Fig. The crank CQ rotates in a clockwise direction with a constant speed of 300 r.p.m. Calculate the velocity of the piston P at the instant when the angle $\theta = 30^\circ$. Solve it by graphical method.



Analysis of Four Bar Mechanism By Instantaneous Centre Method:

This four bar mechanism, which consists of a fixed link AD , two movable links AB and CD rotating about points A and D respectively, and a connecting link BC (which is also known as coupler BC). Let the link AB is rotating at a uniform angular velocity, and it is required to find the corresponding motions of the two links BC and CD .



ω_{AB} = Angular velocity of link AB , rotating about A in the clock-wise direction. The value of ω_{AB} is given as ω_1 .

ω_{CD} = Angular velocity of link CD , rotating about D . The value of ω_{CD} is to be calculated which is ω_2 .

ω_{BC} = Angular velocity of the link BC . The value of ω_{BC} is also to be calculated.

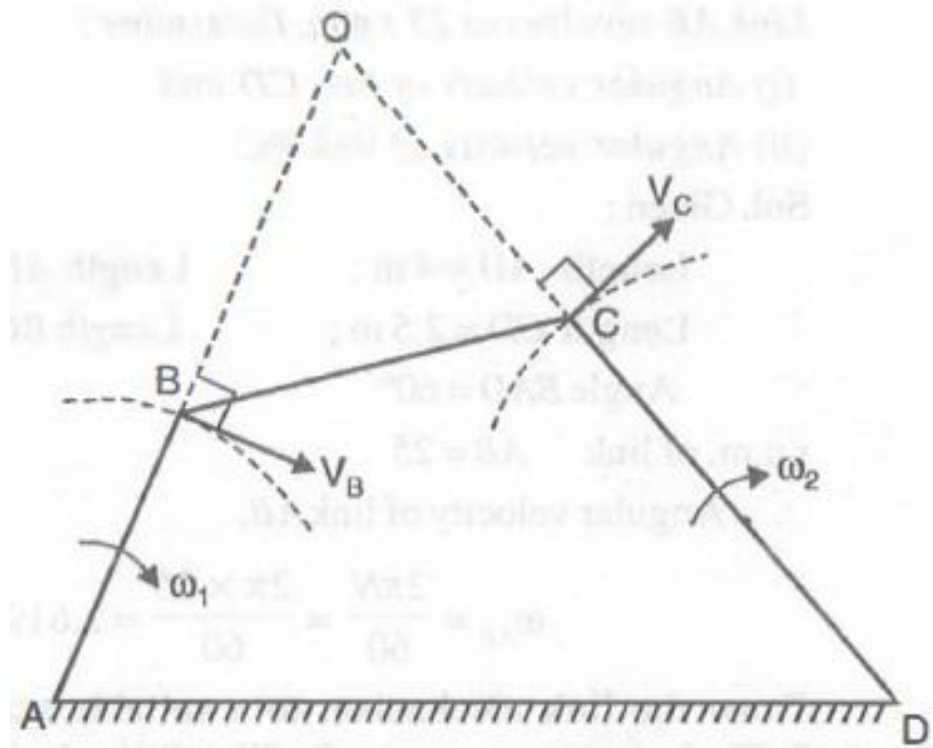
V_B = Linear velocity of point B in the direction perpendicular to AB .

$$= \omega_{AB} \times AB$$

V_C = Linear velocity of point C in the direction perpendicular to CD .

$$= \omega_{CD} \times CD$$

- The link AB and link CD are having motion of rotation, whereas the link BC is having motion of translation as well as rotation. The instantaneous centre of link BC for the given position is determined by drawing normal to the directions of velocity V_B and V_C .
- The normal to the direction V_B is the line AB whereas the normal to the direction V_C is line CD . Hence produce line AB and CD . The intersection of these lines gives the point O (i.e., instantaneous centre) for the link BC .



- The linear velocity at B is given by $V_B = \omega_{BC} \times BO$
- The linear velocity at C is given by $V_C = \omega_{BC} \times CO$

Equating the linear velocity at V_B

$$\omega_{AB} \times AB = \omega_{BC} \times BO$$

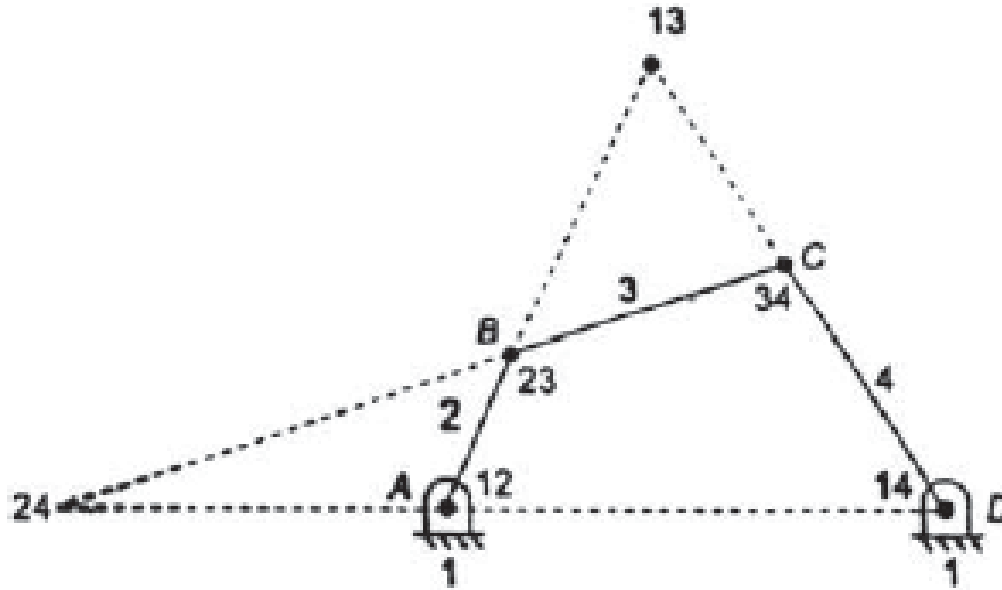
$$\omega_{BC} = \frac{\omega_{AB} \times AB}{BO}$$

Similarly equating the liner velocity at C:

$$\omega_{CD} \times CD = \omega_{BC} \times CO$$

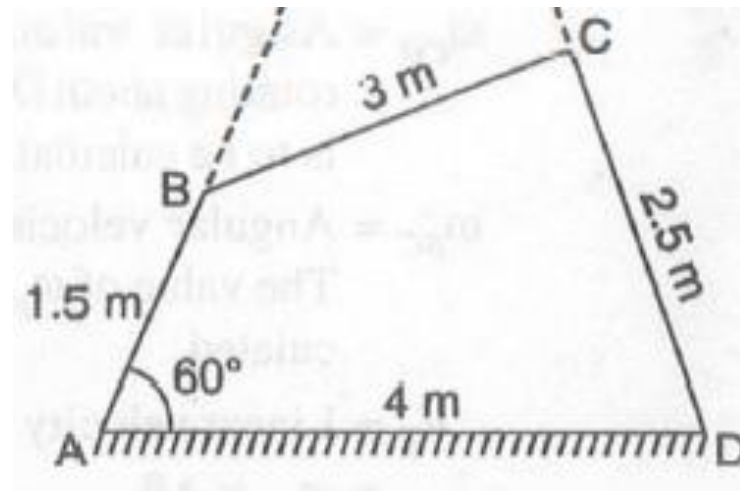
$$\omega_{CD} = \frac{\omega_{BC} \times CO}{CD}$$

Possible ICs for four bar mechanism:



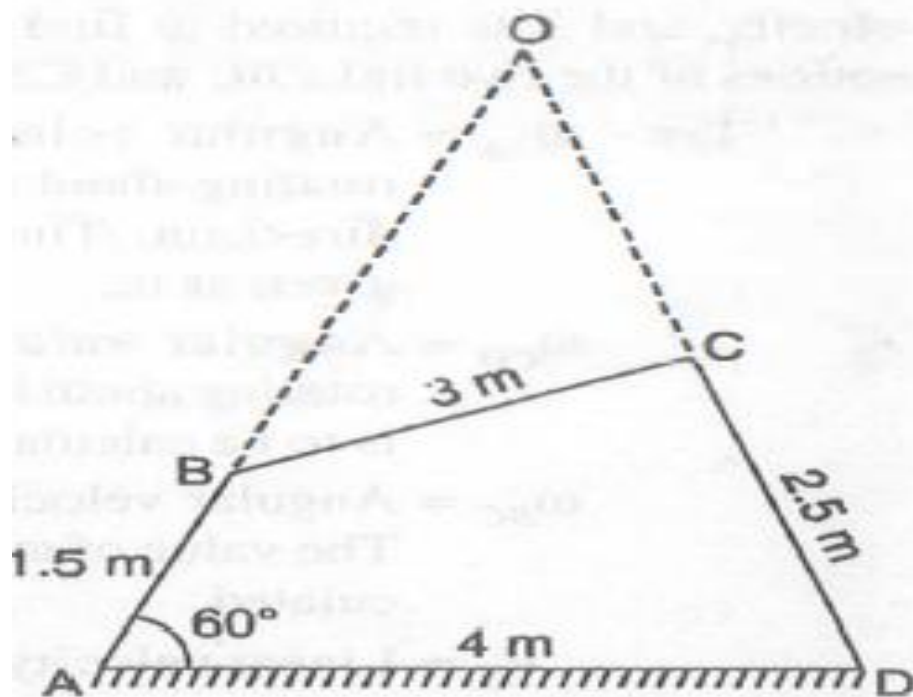
Q 1: The Fig shows a pin-jointed four bar linkage having the following dimensions : Fixed link $AD=4\text{m}$; Driving link $AB=1.5\text{ m}$ Driven link $CD=2.5\text{m}$; Connecting link $BC=3\text{m}$, Angle $BAD=60^\circ$. Link AB revolves at 25 r.p.m. Determine :

- (i) Angular velocity of link CD and
- (ii) Angular velocity of link BC .



- Length $AD=4\text{m}$; Length $CD=2.5\text{m}$;
- Angle $BAD=60^\circ$, r.p.m. of link $AB=25$
- Angular velocity of link AB - $\omega_{AB} = \frac{2\pi N}{60} = \frac{2\pi \times 25}{60} = 2.619 \text{ rad/s.}$

- Draw the link mechanism to a suitable scale as shown in Fig. The instantaneous centre O of link BC is obtained by producing line AB and DC . Now measure the distance CO and BO from



- By measurement, $CO = 3 \text{ m}$ and $BO = 4.65 \text{ m}$.
- First calculate "the angular velocity of link BC.
- Let ω_{BC} = angular velocity of link BC.

$$\omega_{BC} = \frac{\omega_{AB} \times AB}{BO} = \frac{2.619 \times 1.5}{4.65} \text{ rad/s}$$
$$= 0.845 \text{ rad/s. Ans.}$$

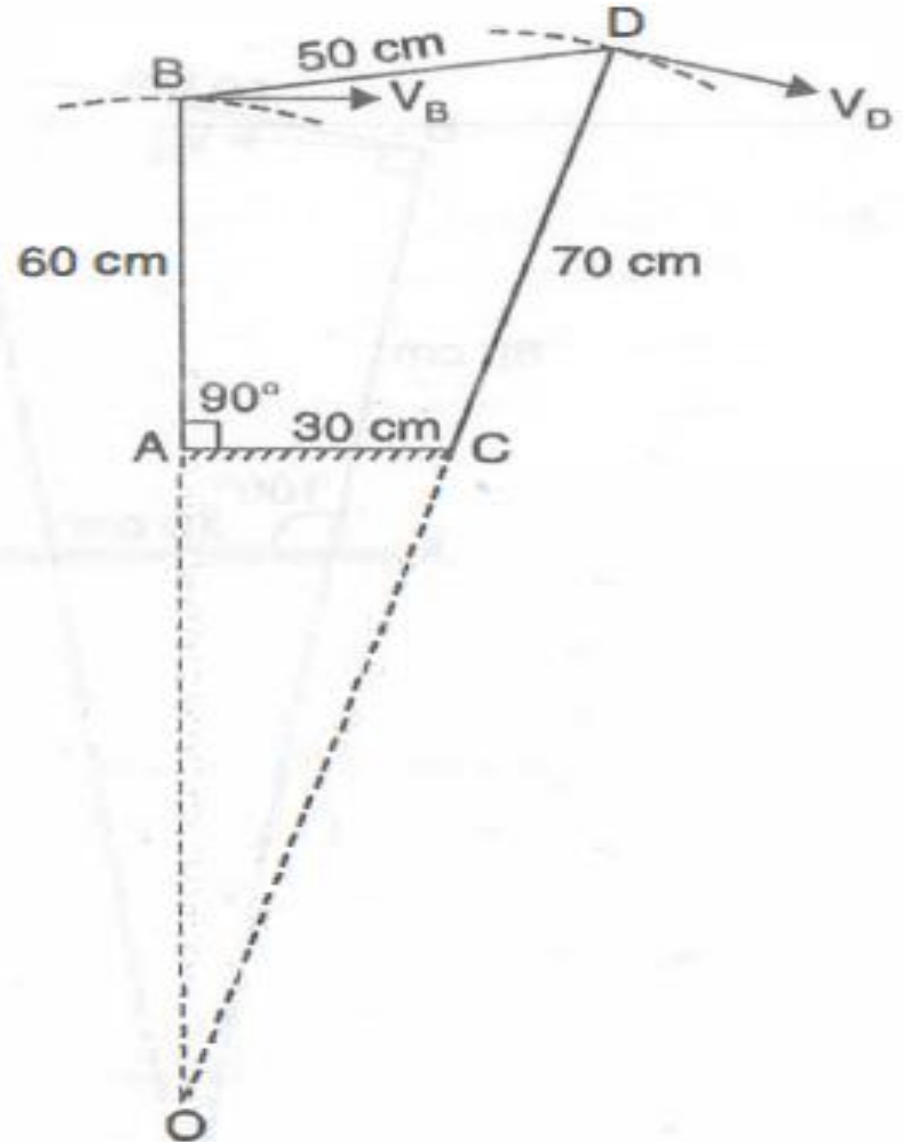
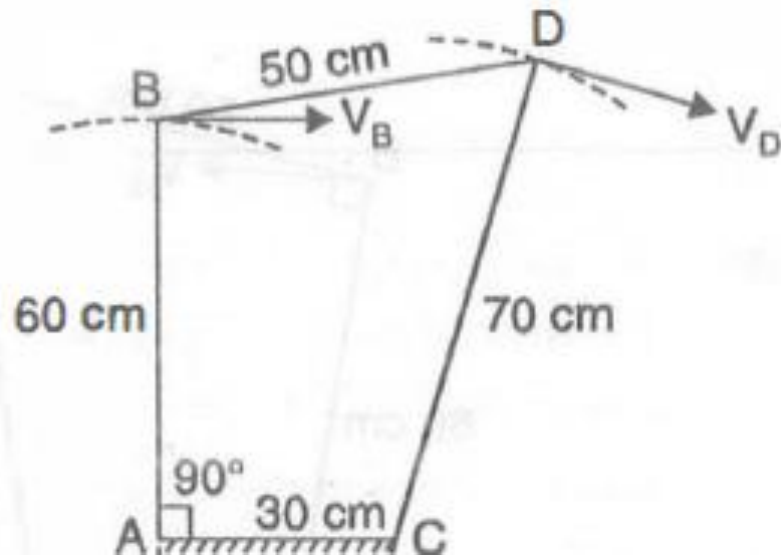
$$\omega_{CD} = \frac{\omega_{BC} \times CO}{CD} = \frac{0.845 \times 3.0}{2.5} = 1.014 \text{ rad/s.}$$

Q 2: A four bar mechanism ABCD, points A and C are fixed points 30 cm apart and AB, CD are bars 60 cm and 70 cm long respectively, which are connected by a rod BD which is 50 cm long. If AB rotates with uniform speed of 60 r.p.m., Determine :

- (i) Velocity of D when AB is perpendicular to AC and also when it makes 10° on either side of the perpendicular*
- (ii) The instantaneous centre of the bar BD and its angular velocities in the three positions.*

- Ans: Length of fixed link AC=30 cm*
- Length of link AB=60cm*
- Length of link CD=70cm*
- Length of link BD=50 cm*
- Speed of link AB=60r.p.m*

Draw the FBD:



By measurement,

$$OC = 107 \text{ cm and } AO = 103 \text{ cm}$$

$$OD = OC + CD = 107 + 70 = 177 \text{ cm}$$

$$OB = OA + AB = 103 + 60 = 163 \text{ cm}$$

$$\begin{aligned} V_B &= \omega_{AB} \times AB = 2\pi \times 60 \text{ cm/s} \\ &= 376.8 \text{ cm/s} = 3.768 \text{ m/s.} \end{aligned}$$

Angular velocity of link AB ,

$$\omega_{AB} = \frac{2\pi N}{60} = \frac{2\pi \times 60}{60} = 2\pi \text{ rad/s.}$$

The velocity V_B is also equal to $\omega_{BD} \times OB$ where ω_{BD} = angular velocity of point B or of rod BD about O .

$$\therefore V_B = \omega_{BD} \times OB = \omega_{BD} \times \frac{163}{100} \text{ m/s}$$

Equating the two values of V_B ,

$$3.768 = \omega_{BD} \times \frac{163}{100}$$

$$\therefore \omega_{BD} = \frac{3.768 \times 100}{163} = 2.3116 \text{ rad/s.}$$

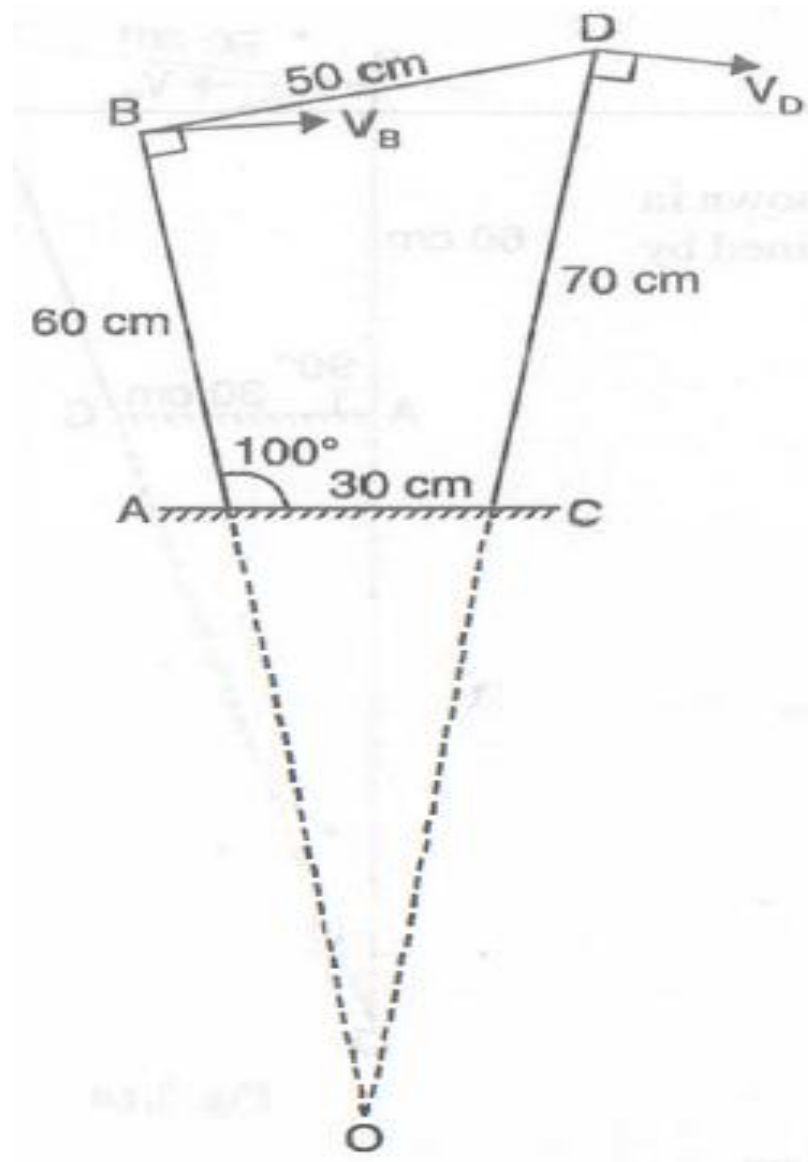
The velocity V_D at point D is given by

$$V_D = \omega_{BD} \times OD$$

$$= 2.3116 \times \frac{177}{100}$$

$$= 4.09 \text{ m/s. Ans.}$$

Second Case: when it makes 10° on either side of the perpendicular
i.e. When the angle between AB and AC = 100° .



By measurement:

$$AO = 98 \text{ cm and } OC = 97 \text{ cm}$$

$$OB = OA + AB = 98 + 60 = 158 \text{ cm} = 1.58 \text{ m}$$

$$OD = OC + CD = 97 + 70 = 167 \text{ cm} = 1.67 \text{ m}$$

$$V_B = \omega_{AB} \times AB$$

With A as centre of rotation

$$= \omega_{BD} \times OB$$

With O as centre of rotation

$$\omega_{BD} \times OB = \omega_{AB} \times AB$$

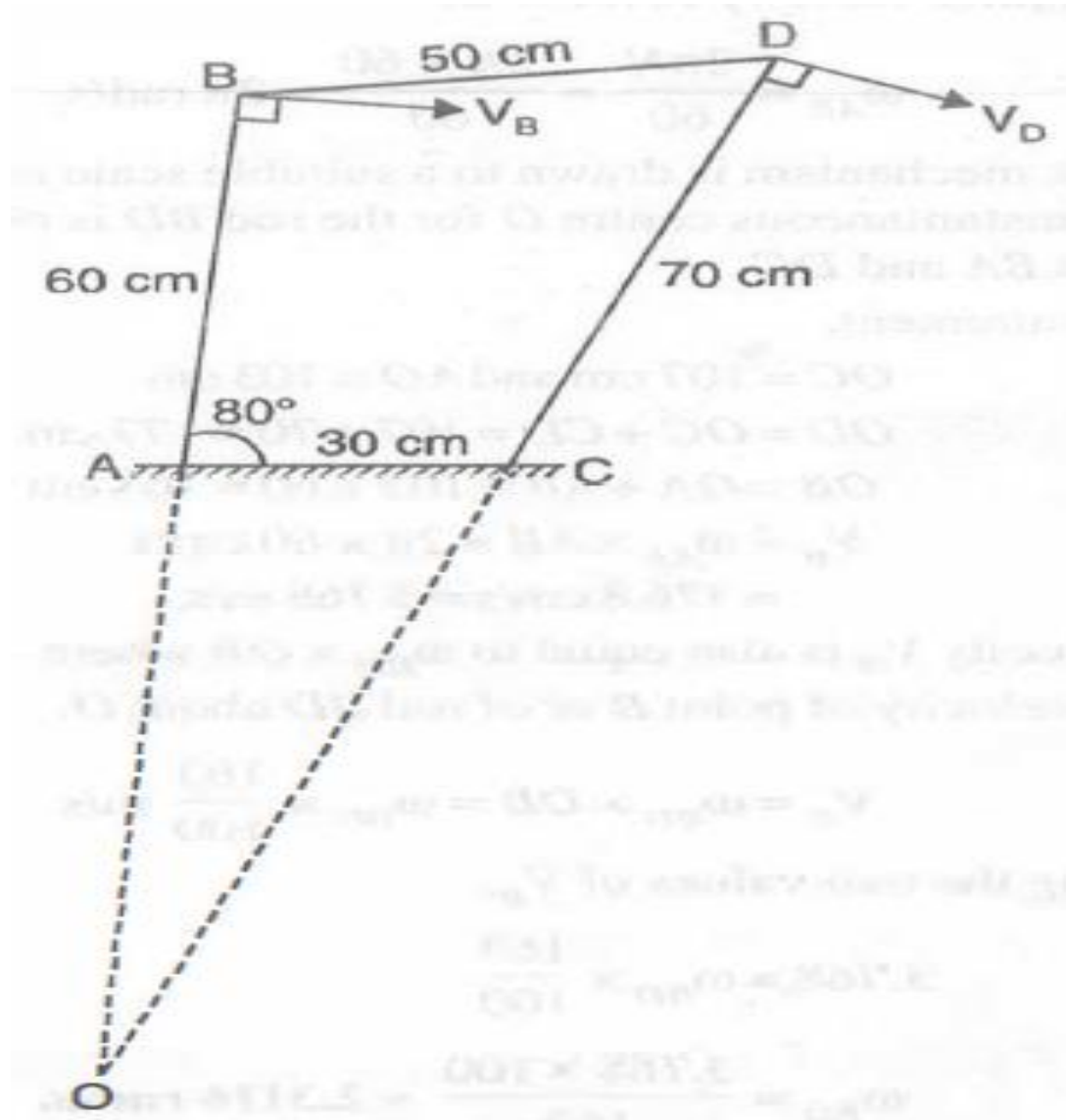
$$\omega_{BD} = \frac{\omega_{AB} \times AB}{OB} = \frac{2\pi \times 60}{158} = 2.386 \text{ rad/s. Ans.}$$

$$V_D = \omega_{BD} \times OD$$

with O as centre of rotation

$$= 2.386 \times 1.67 = 3.984 \text{ m/s. Ans.}$$

Third Case: when it makes 10° on either side of the perpendicular i.e.
When the angle between AB and AC = 80° .



By measurement: $AO = 100 \text{ cm}$ and $OC = 109 \text{ cm}$

$$OB = OA + AB = 100 + 60 = 160 \text{ cm or } 1.6 \text{ m}$$

$$OD = OC + CD = 109 + 70 = 179 \text{ cm or } 1.79 \text{ m}$$

$$V_B = \omega_{AB} \times AB$$

with A as centre of rotation

$$= \omega_{BD} \times OB$$

with O as centre of rotation

$$\omega_{BD} \times OB = \omega_{AB} \times AB$$

$$\omega_{BD} = \frac{\omega_{AB} \times AB}{OB} = \frac{2\pi \times 60}{160} = 2.356 \text{ rad/s. Ans.}$$

$$V_D = \omega_{BD} \times OD$$

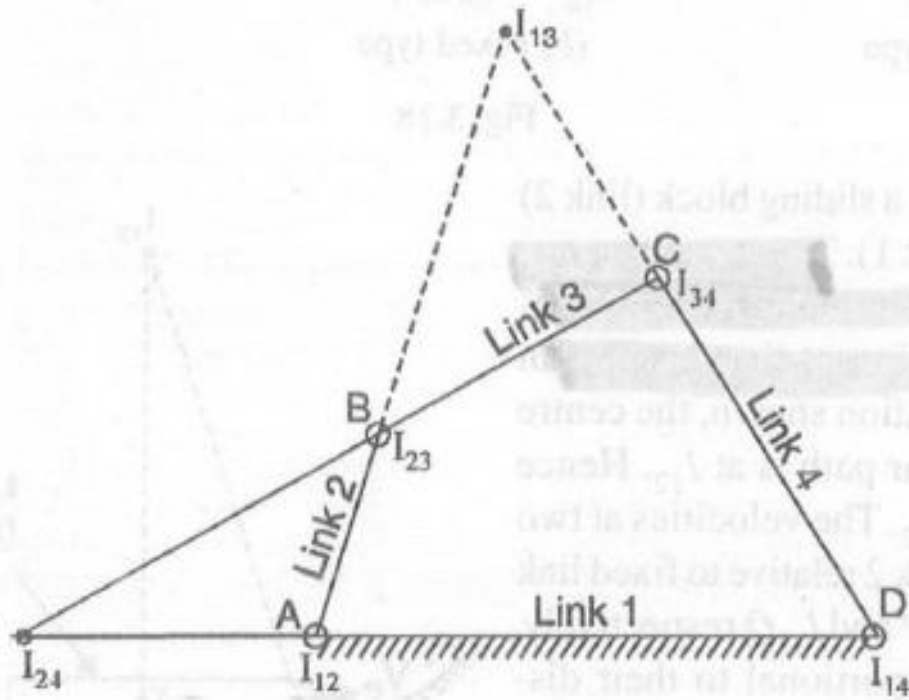
with O as centre of rotation

$$= 2.356 \times 1.79 = 4.217 \text{ m/s. Ans.}$$

Number and Types of instantaneous Centres in a Mechanism:

$$N = \frac{n(n-1)}{2}$$

$$N = \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6.$$



- *Kennedy Theorem (Three Centres-in-line Theory)*
- *It states that if three kinematic links move relatively to each other, their instantaneous centers.*

<i>Links</i>	1	2	3	4
<i>Instantaneous Centres</i>	12 13 14	23 24	34	—